# <u>Minor Project 1</u>

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# Diabetes Patient Risk Prediction Using Supervised Machine Learning


## *Problem Statement*

Diabetes is a chronic disease affecting millions worldwide. Early prediction of diabetes risk can help in timely medical intervention. The goal of this project is to build a **supervised machine learning model that predicts whether a patient is diabetic or not**, based on diagnostic health measurements.

## *Dataset Description*

- **Name:** Pima Indians Diabetes Database
- **Source:** Kaggle
- **Link:** https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database
- **Rows:** 768 | **Columns:** 9
- **Features:** Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age

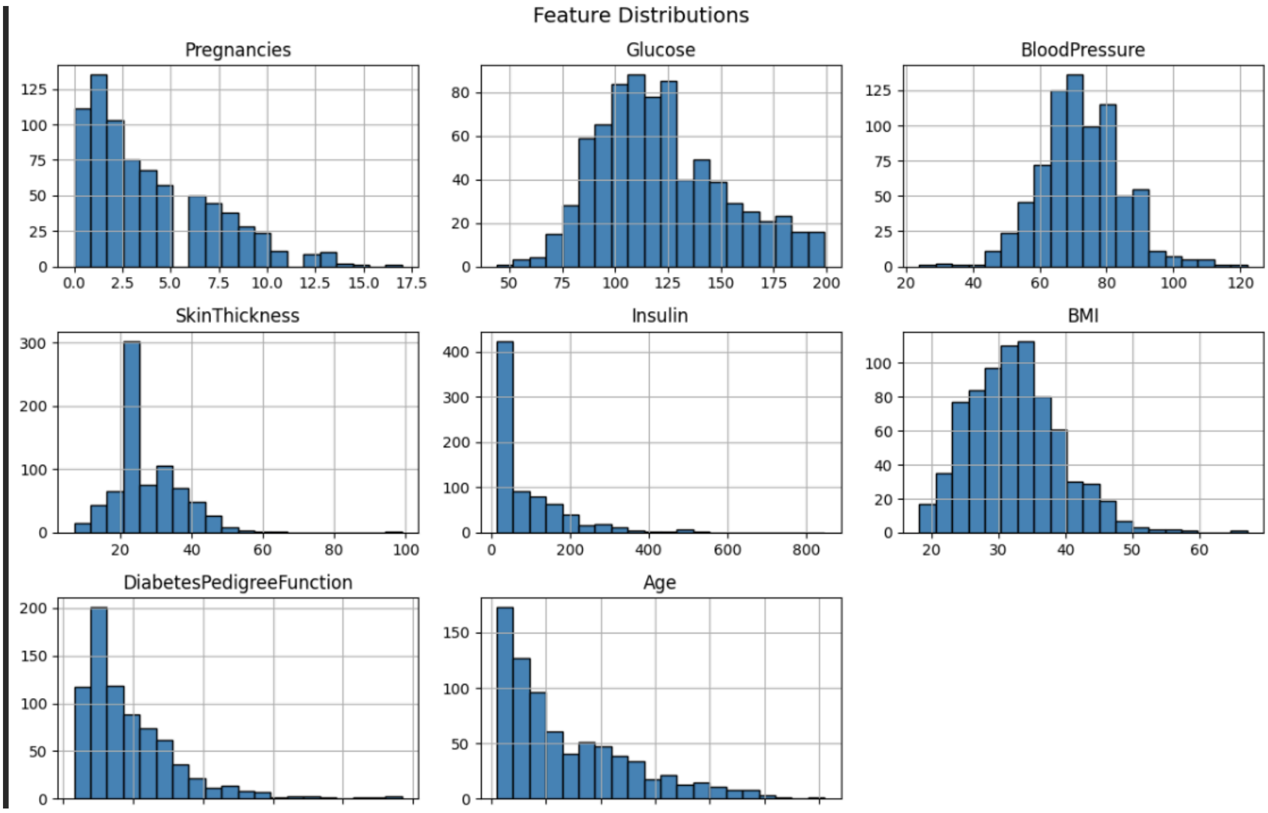
- **Target:** Outcome (0 = Non-Diabetic, 1 = Diabetic)

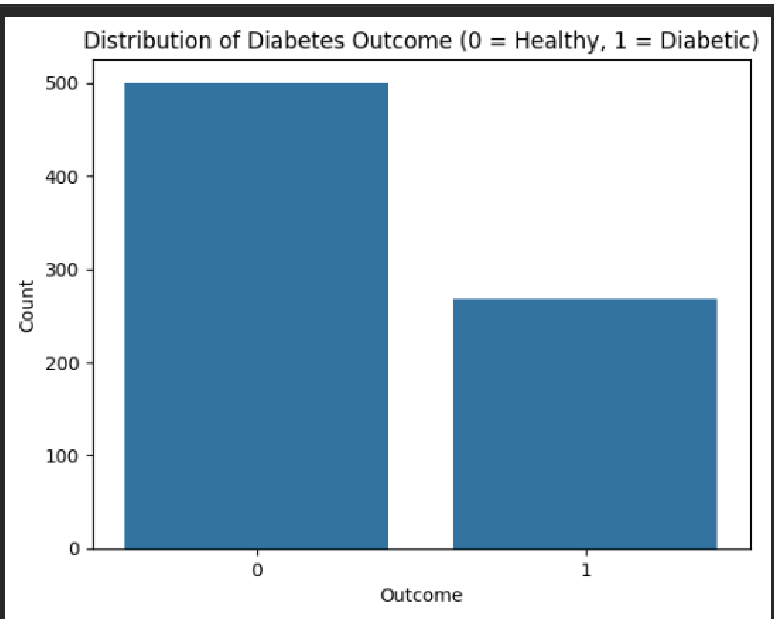
## *4. Data Preprocessing Steps*

- Identified that columns like Glucose, BloodPressure, SkinThickness, Insulin, and BMI contained zero values, which are medically impossible. These were replaced with the **median** of their respective columns.
- Checked for duplicate rows — none found.
- Checked for null/missing values — none found after zero treatment.
- Applied **StandardScaler** to normalize features for Logistic Regression.
- Split data into **80% training and 20% testing** using `train_test_split` with `random_state=42`.

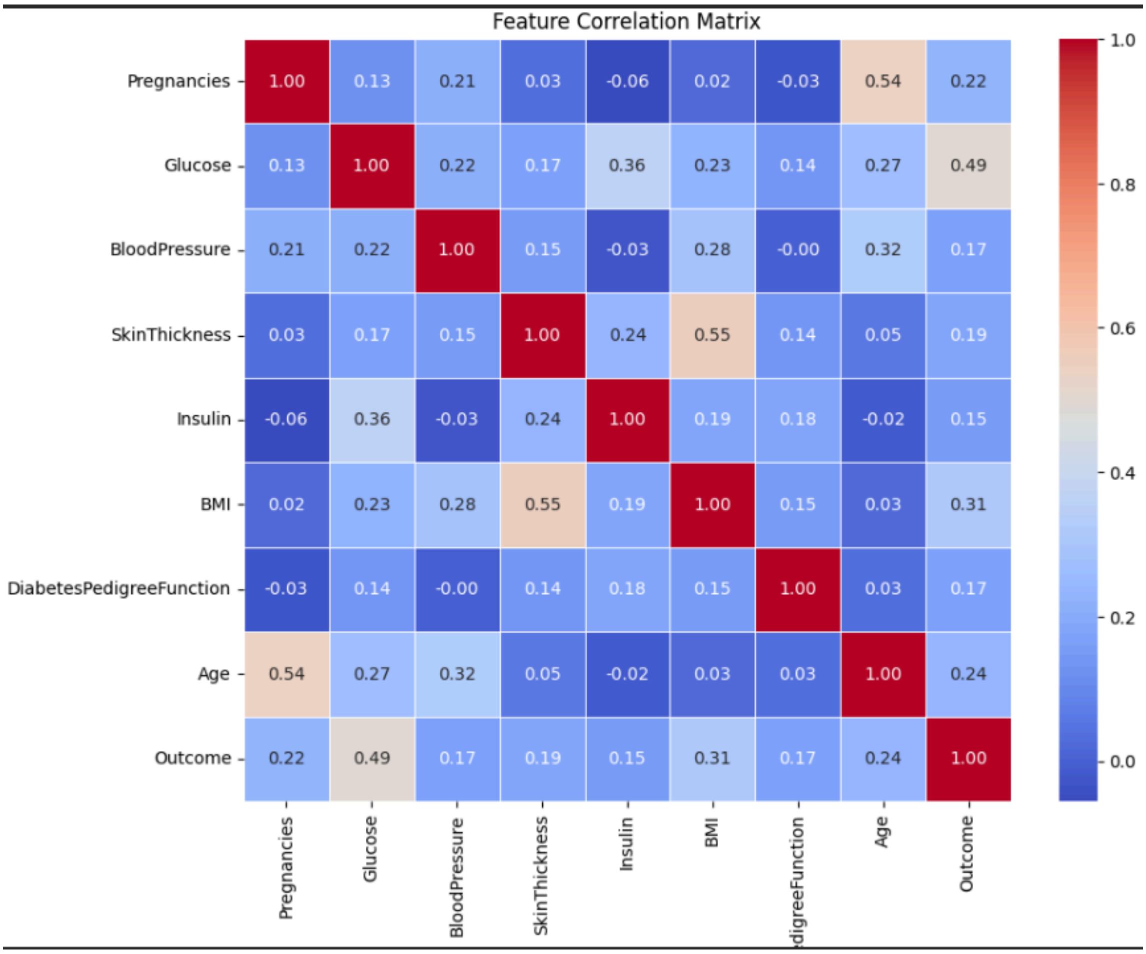
## *5. Exploratory Data Analysis*

- The dataset is slightly imbalanced: 500 non-diabetic vs 268 diabetic patients.
- **Glucose** and **BMI** show strong positive correlation with the Outcome.
- Histograms revealed that most features are right-skewed.
- Correlation heatmap showed Glucose has the highest correlation with Outcome (~0.49).
- Feature distributions were visualized using histograms and a heatmap.

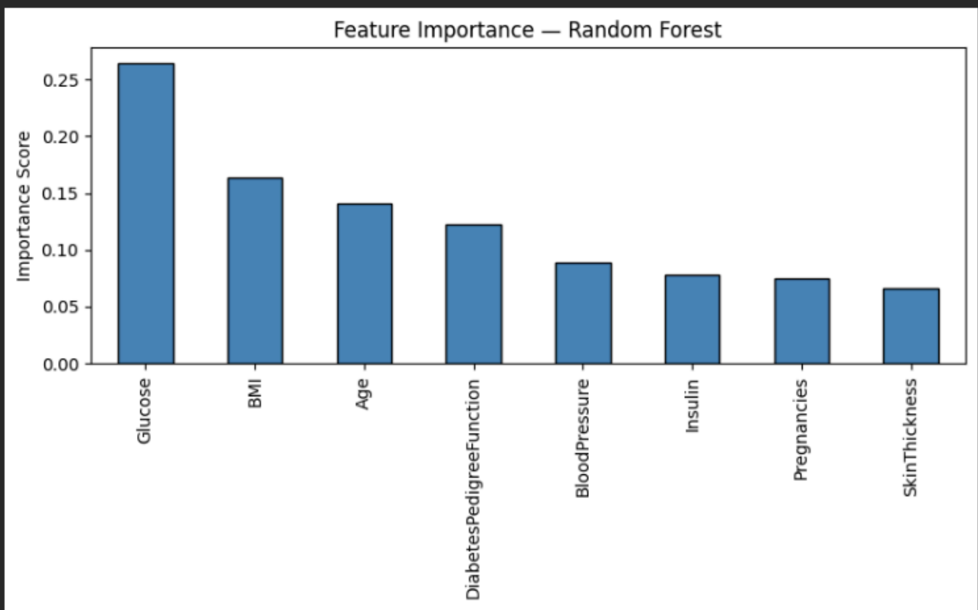



Here it shows roughly 500 people are non-diabetic and approx. 250 people are diabetic , there is a slight class imbalance.




```
print("Top features:\n", feature_imp)
```




```
Top features:
 Glucose                    0.264943
BMI                         0.163291
Age                         0.141010
DiabetesPedigreeFunction    0.122428
BloodPressure               0.089150
Insulin                     0.077839
Pregnancies                 0.074988
SkinThickness               0.066351
dtype: float64
```

## ***6. Model Used***

Two supervised classification models were trained:

Model 1 — Logistic Regression
A linear model used for binary classification. It estimates the probability of a patient being diabetic using a sigmoid function.

Model 2 — Random Forest Classifier
An ensemble model that builds multiple decision trees and combines their predictions. It handles non-linear relationships well and provides feature importance scores.

## ***7. Training Process***

- Data was split: 80% for training (614 samples), 20% for testing (154 samples).
- Logistic Regression was trained on **scaled** features.
- Random Forest (100 estimators) was trained on **unscaled** features as it is tree-based.
- `random_state=42` was used throughout for reproducibility.

## ***8. Evaluation Metrics and Results***

| Metric | Logistic Regression | Random Forest |
|---|---|---|
| Accuracy | 0.7662 | 0.7532 |
| Precision | 0.6792 | 0.6491 |
| Recall | 0.6545 | 0.6727 |

| | | |
|---|---|---|
| F1 Score | 0.6667 | 0.6607 |

- Confusion Matrix was plotted for both models.
- Random Forest additionally provided **Feature Importance** — Glucose, BMI, and Age were the top 3 predictors.
- Both models performed comparably; Logistic Regression was slightly better in accuracy.

## ***9. Conclusion***

This project successfully demonstrated a complete supervised machine learning workflow for diabetes risk prediction. Two models — Logistic Regression and Random Forest — were trained and evaluated. Glucose level, BMI, and Age were found to be the most significant risk factors. The models achieved approximately 76–77% accuracy. Future improvements could include hyperparameter tuning, handling class imbalance using SMOTE, and trying advanced models like XGBoost.